Predicting Stroke Risk: A Statistical Modeling Approach

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Introduction

Strokes are a major health concern globally, ranking as the second leading cause of death worldwide and fifth leading cause of death in the United States. Stroke occurs when blood flow to the brain is interrupted, either due to a blocked artery (ischemic) or ruptured blood vessel (hemorrhagic). Despite advancements in medical research, understanding and mitigating stroke risk is still a critical public health challenge. Identifying key health factors that are associated with stroke is important for developing prevention strategies and improving patient outcomes.

The primary goal of this study is to develop predictive models to assess the important health factors that may be associated with stroke risk. We aim to use linear regression, model selection techniques, and LASSO regression to determine which health variables are most strongly correlated with stroke occurrence. While some risk factors, such as age and genetics, are unable to be changed, this study focuses on several lifestyle factors that could contribute significantly to stroke risk. By identifying the most important lifestyle and health factors, our study can provide valuable insights that can aid in reducing stroke prevalence and creating effective prevention strategies.

Background

This study leverages publicly available data to analyze the possible key contributing factors with stroke. The data is taken from the Center of Disease Control (CDC) website and is census data collected from the top 500 cities in the United States, last updated in December 2023. By examining trends between these factors, it seeks to identify correlations that can help future research and policymaking. To provide a comprehensive analysis, we want to focus on several health metrics that reflect overall health conditions. These metrics include obesity, smoking, poor physical health, poor mental health, asthma, diabetes, high blood pressure, sleep issues, and high cholesterol.

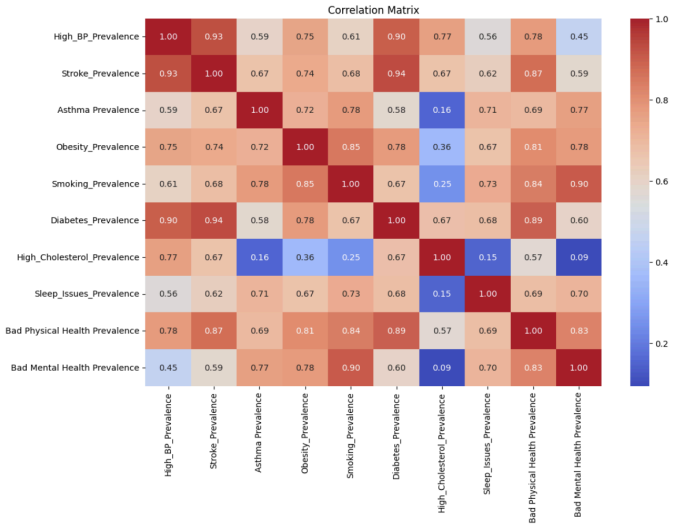
Obesity was identified as a key factor because high body mass index is associated with an increased risk of stroke due to its connection to hypertension and cardiovascular disease.

Smoking was chosen because it damages blood vessels, increases blood pressure, and could form blood clots, all of which significantly increase the risk of stroke. Poor physical health was included because it could indicate a lifestyle that contributes to stroke risk. We chose poor mental health because stress, anxiety, and depression can lead to increased inflammation and higher blood pressure. Asthma was included as a predictor because studies suggest that chronic inflammation and reduced oxygen levels could contribute to cardiovascular complications. Diabetes was chosen because it limits the body's ability to regulate blood sugar, which leads to vascular damage and elevates risk of both ischemic and hemorrhagic strokes. High blood pressure is an established risk factor for stroke, as it weakens blood vessel walls and increases the likelihood of blockages (ischemic) and ruptures (hemorrhagic). Sleep issues were included because sleep disorders have been linked to high blood pressure, irregular heart rhythms, and reduced oxygen supply to the brain. High cholesterol was chosen because excess cholesterol could result in plaque buildup in arteries which could restrict blood flow and increase stroke risk.

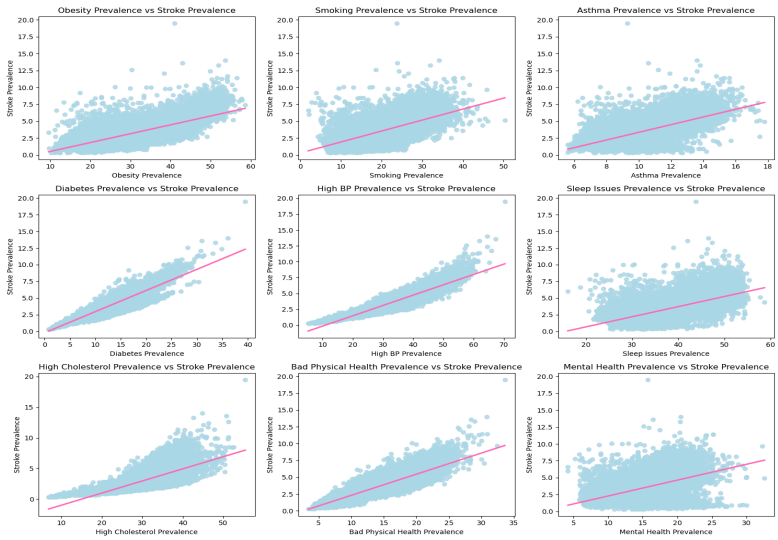
Exploratory Data Analysis

To better understand our dataset, we conducted exploratory data analysis. Our dataset was cleaned by removing all missing values and assessing the presence of outliers. We generated a correlation heatmap to visualize the relationships between predictor variables. This also helped us to assess the potential of multicollinearity and identify highly correlated variables that may make our model redundant. Next, we conducted simple linear regression for each predictor variable against our response variable. This helps us to determine if any single predictor is particularly effective at predicting stroke risk.

Based on our correlation heatmap, we found that all of our variables have substantial correlations with each other. Our heatmap revealed that stroke and high blood pressure, as well as, stroke and diabetes, have very high correlations of 0.93 and 0.94. We also found that some predictor variables such as diabetes and bad physical health, or, smoking and bad mental health, had high correlations as well.



These results indicate that there might be potential of multicollinearity in our model, which we will take into consideration.



When we plotted all the predictor variables against our response variable, stroke prevalence, they all showed positive linear relationships, emphasizing the strong association between these health risk factors and the likelihood of stroke occurrence.

Methodology

To develop a robust model for predicting stroke risk, we used a variety of statistical techniques aimed at understanding the relationships between our chosen predictor variables and stroke. Our process includes Exploratory Data Analysis (EDA), regression modeling, dimensionality reduction, and feature selection.

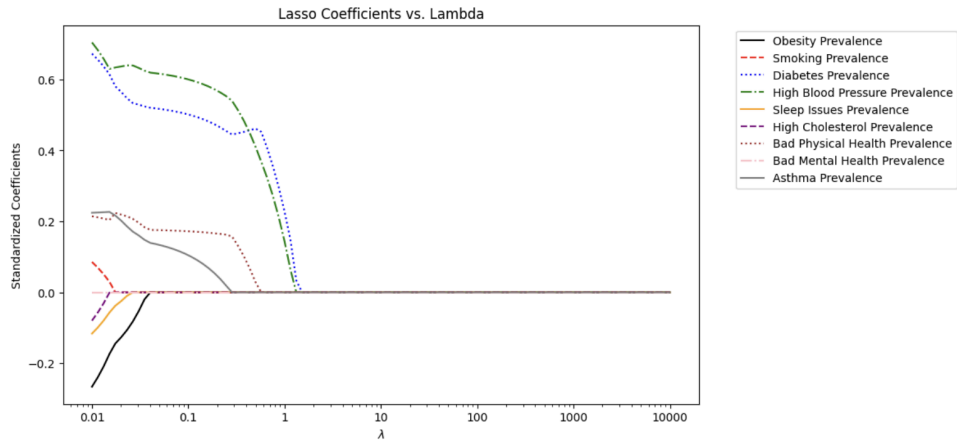
Following our exploratory data analysis, we performed multiple linear regression using all the predictor variables to assess their combined influence on stroke risk. This allowed us to determine which predictors were statistically significant when predicting stroke risk in a model containing all predictors.

We applied Principal Component Analysis (PCA) to reduce dimensionality while still keeping the important information. This transformed our correlated variables into a set of uncorrelated principal components, helping us improve model interpretability.

To further refine our model, we implemented LASSO regression, a regularization technique that effectively removed the predictors that were not significant. This prevents overfitting and selects the important features to remain in the model. We also conducted both forward and backward

stepwise selection to identify the best subset of predictor variables that would result in the best predictive model.

Main Results

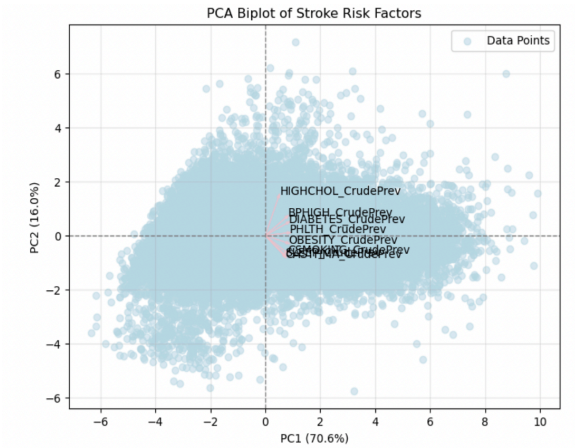


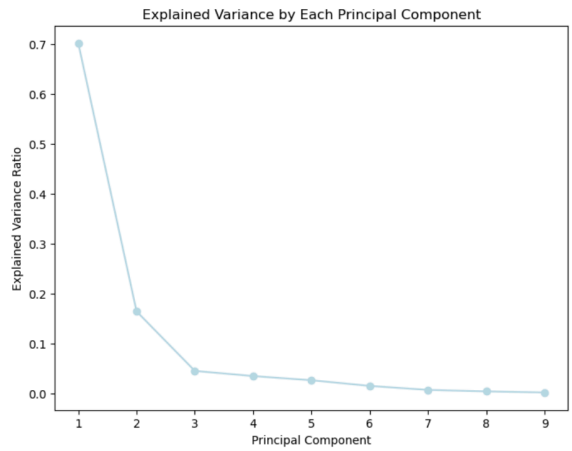
Based on our lasso regression analysis, there is further support of the key predictors for stroke risk. The LASSO coefficient plot shows that diabetes and high blood pressure remain as the last two nonzero coefficients as **λ** gets larger, indicating that these variables are the strongest predictors of stroke risk. As **λ** increases, other predictor variables shrink to zero. These findings highlight the importance of targeting diabetes and high blood pressure management in stroke prevention strategies.

To view the combined influence of our predictor variables on stroke influence, we performed a multiple linear regression analysis. This allowed us to evaluate the contribution of each variable while making sure the effects of other variables were taken into account. Each predictor variable had a p-value of <0.000, indicating that each predictor is significant in predicting stroke risk. From this analysis, we can see that blood pressure has a regression coefficient of 0.0955, asthma 0.151, obesity -0.0489, smoking 0.043, diabetes 0.1691, high cholesterol -0.0653, sleep issues -0.0386, poor physical health 0.1306, and poor mental health -0.0863. Significantly, blood pressure, asthma, diabetes, and poor physical health have the strongest effects on predicting stroke.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:       STROKE_CrudePrev   R-squared:                       0.953
Model:                            OLS   Adj. R-squared:                  0.953
Method:                 Least Squares   F-statistic:                 6.101e+04
Date:                Mon, 10 Mar 2025   Prob (F-statistic):               0.00
Time:                        15:00:21   Log-Likelihood:                -7169.4
No. Observations:               27210   AIC:                         1.436e+04
Df Residuals:                   27200   BIC:                         1.444e+04
Df Model:                           9
Covariance Type:            nonrobust
=========================================================================================
                            coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------------
const                     0.6833      0.036     19.179      0.000       0.613       0.753
BPHIGH_CrudePrev          0.0955      0.001    107.808      0.000       0.094       0.097
CASTHMA_CrudePrev         0.1510      0.002     65.163      0.000       0.146       0.156
OBESITY_CrudePrev        -0.0489      0.001    -81.102      0.000      -0.050      -0.048
CSMOKING_CrudePrev        0.0430      0.001     42.948      0.000       0.041       0.045
DIABETES_CrudePrev        0.1691      0.002     84.578      0.000       0.165       0.173
HIGHCHOL_CrudePrev       -0.0653      0.001    -61.628      0.000      -0.067      -0.063
SLEEP_CrudePrev          -0.0386      0.001    -60.972      0.000      -0.040      -0.037
PHLTH_CrudePrev           0.1306      0.003     48.379      0.000       0.125       0.136
MHLTH_CrudePrev          -0.0863      0.002    -38.480      0.000      -0.091      -0.082
==============================================================================
Omnibus:                     9189.988   Durbin-Watson:                   2.003
Prob(Omnibus):                  0.000   Jarque-Bera (JB):           223886.120
Skew:                           1.072   Prob(JB):                         0.00
Kurtosis:                      16.888   Cond. No.                     1.36e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.36e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
Mean Squared Error: 0.10875126988348488
R-squared: 0.9499223928245386
```





After conducting Principal Component Analysis, it was found that three principal components were able to explain over 90% of the variance in the data. We also produced a PCA biplot of the first two principal components, which shows that PC1 captures 70.6% of the variance in our dataset, whereas PC2 captures 16% of the variance. This analysis shows that most of the structure in our dataset could be represented in a lower-dimensional space, which could result in a model that generalizes better and does not overfit.

```
Selected Features: ['BPHIGH_CrudePrev', 'CASTHMA_CrudePrev', 'OBESITY_CrudePrev', 'CSMOKING_CrudePrev', 'DIABETES_CrudePrev', 'HIGHCHOL_Crude
Prev', 'SLEEP_CrudePrev', 'PHLTH_CrudePrev', 'MHLTH_CrudePrev']
                            OLS Regression Results
==============================================================================
Dep. Variable:       STROKE_CrudePrev   R-squared:                       0.953
Model:                            OLS   Adj. R-squared:                  0.953
Method:                 Least Squares   F-statistic:                 6.101e+04
Date:                Mon, 10 Mar 2025   Prob (F-statistic):               0.00
Time:                        14:53:48   Log-Likelihood:                -7169.4
No. Observations:               27210   AIC:                         1.436e+04
Df Residuals:                   27200   BIC:                         1.444e+04
Df Model:                           9
Covariance Type:            nonrobust
=======================================================================================
                          coef    std err          t      P>|t|      [0.025      0.975]
---------------------------------------------------------------------------------------
const                   0.6833      0.036     19.179      0.000       0.613       0.753
BPHIGH_CrudePrev        0.0955      0.001    107.808      0.000       0.094       0.097
CASTHMA_CrudePrev       0.1510      0.002     65.163      0.000       0.146       0.156
OBESITY_CrudePrev      -0.0489      0.001    -81.102      0.000      -0.050      -0.048
CSMOKING_CrudePrev      0.0430      0.001     42.948      0.000       0.041       0.045
DIABETES_CrudePrev      0.1691      0.002     84.578      0.000       0.165       0.173
HIGHCHOL_CrudePrev     -0.0653      0.001    -61.628      0.000      -0.067      -0.063
SLEEP_CrudePrev        -0.0386      0.001    -60.972      0.000      -0.040      -0.037
PHLTH_CrudePrev         0.1306      0.003     48.379      0.000       0.125       0.136
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==============================================================================
Omnibus:                     9189.988   Durbin-Watson:                   2.003
Prob(Omnibus):                  0.000   Jarque-Bera (JB):           223886.120
Skew:                           1.072   Prob(JB):                         0.00
Kurtosis:                      16.888   Cond. No.                     1.36e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.36e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
```

Conducting both forward and backward stepwise selection resulted in the same final model chosen, with all predictors included. This means that each predictor contributed significantly to explaining the variance in the outcome, showing that each of the health predictors we evaluated is relevant for predicting stroke risk. The adjusted R-squared for this final model was 0.953,

meaning that 95.3% of the variance in the variable for stroke occurrence could be explained by the predictors in the model: obesity, smoking, poor physical health, poor mental health, asthma, diabetes, high blood pressure, sleep issues, and high cholesterol. This high adjusted R-squared value indicates that our model fits well without being too complex.

Discussion & Outlook

One of the strengths of this study is the use of different statistical modeling techniques which lets us make a variety of comparisons and helps identify the most important predictors while minimizing the risk of overfitting. Also, the use of data from a reputable source, the Centers for Disease Control and Prevention (CDC), ensures the credibility of the study.

However, there are also several limitations to our study that should also be acknowledged. For example, the dataset we used only covers 500 U.S. cities, which may not fully capture the diversity of stroke risk factors throughout the whole country. Furthermore, the data was last updated in 2023, so changes in population health trends since then are not being considered. We should also address that while our study focused on specific health-related variables, there can be other important factors, such as socioeconomic status, diet, exercise habits, and access to healthcare, which were not included in our analysis due to the availability of data. These factors can also play a critical role in stroke risk and should be considered in future research.

Another limitation is the risk of confounding variables and any interactions that failed to get measured between predictors. For instance, while we found strong correlations between stroke risk and high blood pressure, it is possible that other underlying factors, such as medication use, diet, or genetics, can influence this relationship. Our findings also suggested a connection between poor mental health and smoking, but it is unclear whether mental health directly contributes to stroke risk or if it acts as a mediator through behaviors like smoking and poor physical health.

To improve this study, we could include additional variables by merging more datasets that could provide a greater detailed analysis of other potential risk factors. Examining interactions between predictors, such as obesity and diabetes, can reveal how multiple risk factors jointly influence stroke risk and improve the model's predictive accuracy. Additionally, we could use other machine learning techniques like tree based methods to also improve predictive accuracy and

capture complex nonlinear relationships between variables. Future research could further our findings by conducting a prospective study to examine stroke risk factors over time. This would provide a more in-depth analysis of stroke risk and allow for better generalization of the results.

Conclusion

After testing multiple regression models and selection techniques, we identified diabetes and high blood pressure as the most influential predictors of stroke risk, which agreed with existing medical research. This was especially supported by conducting LASSO regression. Simple linear regression showed that each predictor explored had a significant relationship with the variable of stroke. Using forward and backward stepwise selection, the best model found for both methods included all of the predictor variables. Since this model was supported by multiple methods, we decided that the best model to predict stroke risk would contain all predictors: obesity, smoking, poor physical health, poor mental health, asthma, diabetes, high blood pressure, sleep issues, and high cholesterol. This final model explains 95.3% of the variance in stroke risk, which shows its effectiveness in identifying key risk factors. While all of our chosen predictors significantly contributed to stroke prediction, multicollinearity posed some challenges in interpreting individual coefficients. Since the potential of multicollinearity was further supported by our exploratory data analysis, future efforts can take this into consideration. For a predictive model, all nine health variables would be relevant in predicting stroke risk, but a focus on diabetes and high blood pressure would be most effective since they hold the greatest influence.

These findings help us realize that we need better public health efforts to manage diabetes and high blood pressure, which could help lower stroke rates. Future research could add to these prediction models by considering factors like age, socioeconomic status, and lifestyle habits. They could also incorporate other statistical methods, like tree-based models, to reveal more complex patterns in stroke risk factors. By including larger and more diverse datasets we could gain more insights into improving stroke prevention strategies and creating more effective public health policies.

Code Appendix

```
# correlation heatmap
corr_matrix = df.corr()
plt.figure(figsize=(12, 8))
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Matrix")
plt.show()
```

```
#Ordinary Least Square Regression Results

df = df.rename(columns={
    "STROKE_CrudePrev": "Stroke Prevalence",
    "OBESITY_CrudePrev": "Obesity Prevalence",
    "CSMOKING_CrudePrev": "Smoking Prevalence",
    "CASTHMA_CrudePrev": "Asthma Prevalence",
    "DIABETES_CrudePrev": "Diabetes Prevalence",
    "BPHIGH_CrudePrev": "High BP Prevalence",
    "SLEEP_CrudePrev": "Sleep Issues Prevalence",
    "HIGHCHOL_CrudePrev": "High Cholesterol Prevalence",
    "PHLTH_CrudePrev": "Bad Physical Health Prevalence",
    "MHLTH_CrudePrev": "Mental Health Prevalence"
})

predictors = [
    'Obesity Prevalence', 'Smoking Prevalence', 'Asthma Prevalence',
    'Diabetes Prevalence', 'High BP Prevalence', 'Sleep Issues Prevalence',
    'High Cholesterol Prevalence', 'Bad Physical Health Prevalence', 'Mental Health Prevalence'
]

response = 'Stroke Prevalence'
results = {}

for predictor in predictors:
    X = df[predictor]
    Y = df[response]
    X = sm.add_constant(X)
    model = sm.OLS(Y, X).fit()
    results[predictor] = model.summary()

for predictor, result in results.items():
    print(f"Results for {predictor}:")
    print(result)
    print("\n" + "="*80 + "\n")
```

```
#Simple linear regression
fig, axes = plt.subplots(3, 3, figsize=(15, 12))
axes = axes.flatten()

for i, predictor in enumerate(predictors):
    sns.regplot(x=predictor, y=target, data=df,
                scatter_kws={"color": "lightblue"},
                line_kws={"color": "hotpink"},
                ax=axes[i])
    axes[i].set_title(f"{predictor} vs {target}")
    axes[i].set_xlabel(predictor)
    axes[i].set_ylabel(target)

plt.tight_layout()
plt.show()
```

```
#PCA
X = df[predictors]
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
pca = PCA()

pca.fit(X_scaled)
plt.figure(figsize=(8, 6))
plt.plot(range(1, len(pca.explained_variance_ratio_) + 1), pca.explained_variance_ratio_, marker='o', color='lightbl
plt.title('Explained Variance by Each Principal Component')
plt.xlabel('Principal Component')
plt.ylabel('Explained Variance Ratio')
plt.show()
```

```
#PCA Biplot
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

X = df.drop(columns=['STROKE_CrudePrev'])
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
loadings = pca.components_.T
plt.figure(figsize=(8, 6))
plt.scatter(X_pca[:, 0], X_pca[:, 1], alpha=0.5, color="lightblue", label="Data Points")

for i, feature in enumerate(X.columns):
    plt.arrow(0, 0, loadings[i, 0]*2, loadings[i, 1]*2, color="pink", alpha=0.7, head_width=0.1)
    plt.text(loadings[i, 0]*2.2, loadings[i, 1]*2.2, feature, color="black", fontsize=10)

plt.xlabel(f"PC1 ({pca.explained_variance_ratio_[0]*100:.1f}%)")
plt.ylabel(f"PC2 ({pca.explained_variance_ratio_[1]*100:.1f}%)")
plt.title("PCA Biplot of Stroke Risk Factors")
plt.axhline(0, linestyle="--", color="gray", linewidth=1)
plt.axvline(0, linestyle="--", color="gray", linewidth=1)
plt.legend()
plt.grid(alpha=0.3)
plt.show()
```

```
#multiple linear regression
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

file_path = "500_Cities__Census_Tract-level_Data__GIS_Friendly_Format___2019_release_20250310 (1).csv"
df = pd.read_csv(file_path)

target = 'STROKE_CrudePrev'
predictors = ['BPHIGH_CrudePrev', 'CASTHMA_CrudePrev', 'OBESITY_CrudePrev',
              'CSMOKING_CrudePrev', 'DIABETES_CrudePrev', 'HIGHCHOL_CrudePrev',
              'SLEEP_CrudePrev', 'PHLTH_CrudePrev', 'MHLTH_CrudePrev']

df = df[[target] + predictors].dropna()
X = df[predictors]
y = df[target]
X_sm = sm.add_constant(X)
model = sm.OLS(y, X_sm).fit()

print(model.summary())

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
lin_reg = LinearRegression()
lin_reg.fit(X_train, y_train)
y_pred = lin_reg.predict(X_test)

mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print(f"Mean Squared Error: {mse}")
print(f"R-squared: {r2}")
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear model import Lasso
from sklearn.preprocessing import StandardScaler

df = pd.read csv('500 Cities  Census Tract-
level Data  GIS Friendly Format___2019_release_20250310 (1).csv')
target = 'STROKE_CrudePrev'
```

```
features = ['BPHIGH CrudePrev', 'CASTHMA_CrudePrev',
'OBESITY CrudePrev',
              'CSMOKING_CrudePrev', 'DIABETES_CrudePrev',
'HIGHCHOL CrudePrev',
              'SLEEP_CrudePrev', 'PHLTH_CrudePrev', 'MHLTH_CrudePrev']

scaler = StandardScaler()
X scaled = scaler.fit_transform(df[features])
y = df[target]

lambdas = np.logspace(-2, 4, 100)

coefs = []
for lamb in lambdas:
    lasso = Lasso(alpha=lamb, max_iter=10000)
    lasso.fit(X scaled, y)
    coefs.append(lasso.coef_)
coefs = np.array(coefs)

selected vars = features
selected_indices = [features.index(var) for var in selected_vars]

colors = ['black', 'red', 'blue', 'green', 'orange', 'purple',
'brown', 'pink', 'gray']
linestyles = ['solid', 'dashed', 'dotted', 'dashdot', 'solid',
'dashed', 'dotted', 'dashdot', 'solid']

plt.figure(figsize=(10, 6))
plt.xscale("log")
plt.xlabel(r"$\lambda$")
plt.ylabel("Standardized Coefficients")

for i, idx in enumerate(selected indices):
    plt.plot(lambdas, coefs[:, idx], label=selected_vars[i],
color=colors[i], linestyle=linestyles[i])

plt.legend(bbox to anchor=(1.05, 1), loc='upper left')
plt.xticks([1e-2, 1e-1, 1e0, 1e1, 1e2, 1e3, 1e4],
           labels=["0.01", "0.1", "1", "10", "100", "1000", "10000"])
plt.title("Lasso Coefficients vs. Lambda")
plt.show()
```

```python
#forward selection
import pandas as pd
import statsmodels.api as sm

file_path = "500_Cities__Census_Tract-level_Data__GIS_Friendly_Format___2019_release_20250310 (1).csv"
df = pd.read_csv(file_path)

target = 'STROKE_CrudePrev'
predictors = ['BPHIGH_CrudePrev', 'CASTHMA_CrudePrev', 'OBESITY_CrudePrev',
              'CSMOKING_CrudePrev', 'DIABETES_CrudePrev', 'HIGHCHOL_CrudePrev',
              'SLEEP_CrudePrev', 'PHLTH_CrudePrev', 'MHLTH_CrudePrev']

df = df[[target] + predictors].dropna()

def forward_selection(X, y):
    remaining = set(X.columns)
    selected = []
    best_score = float('inf')
    while remaining:
        scores_with_candidates = []
        for candidate in remaining:
            model = sm.OLS(y, sm.add_constant(pd.DataFrame(X[selected + [candidate]]))).fit()
            aic = model.aic
            scores_with_candidates.append((aic, candidate))

        scores_with_candidates.sort()
        best_new_score, best_candidate = scores_with_candidates[0]

        if best_new_score < best_score:
            remaining.remove(best_candidate)
            selected.append(best_candidate)
            best_score = best_new_score
        else:
            break
    return selected

X = df[predictors]
y = df[target]
selected_features = forward_selection(X, y)

final_model = sm.OLS(y, sm.add_constant(X[selected_features])).fit()
print("Selected Features:", selected_features)
print(final_model.summary())
```

```python
#backward selection
file_path = "500_Cities__Census_Tract-level_Data__GIS_Friendly_Format___2019_release_20250310 (1).csv"
df = pd.read_csv(file_path)

target = 'STROKE_CrudePrev'
predictors = ['BPHIGH_CrudePrev', 'CASTHMA_CrudePrev', 'OBESITY_CrudePrev',
              'CSMOKING_CrudePrev', 'DIABETES_CrudePrev', 'HIGHCHOL_CrudePrev',
              'SLEEP_CrudePrev', 'PHLTH_CrudePrev', 'MHLTH_CrudePrev']

df = df[[target] + predictors].dropna()

def backward_selection(X, y):
    selected = list(X.columns)
    best_score = float('inf')

    while len(selected) > 0:
        model = sm.OLS(y, sm.add_constant(X[selected])).fit()
        p_values = model.pvalues[1:]

        max_p_value = p_values.max()

        if max_p_value >= 0.05:
            excluded_feature = p_values.idxmax()
            selected.remove(excluded_feature)
        else:
            break

    return selected

X = df[predictors]
y = df[target]
selected_features = backward_selection(X, y)

final_model = sm.OLS(y, sm.add_constant(X[selected_features])).fit()
print("Selected Features:", selected_features)
print(final_model.summary())
```

References

Berman, R. (2023, October 17). *Death rate from stroke is rising: Lifestyle Can Lower Risk*. Medical News Today. https://www.medicalnewstoday.com/articles/stroke-rates-increasing-50-percent-by-2050

Cepelis, A., Brumpton, B. M., Laugsand, L. E., Langhammer, A., Janszky, I., & Strand, L. B. (2019, December 28). *Asthma, asthma control and risk of ischemic stroke: The hunt study*. Respiratory Medicine: X. https://www.sciencedirect.com/science/article/pii/S2590143519300144

Diabetes & stroke: Causes, symptoms, treatment & prevention. Cleveland Clinic. (2024, December 19). https://my.clevelandclinic.org/health/diseases/9812-diabetes-and-stroke

High cholesterol | stroke association. Stroke Association. (n.d.). https://www.stroke.org.uk/stroke/managing-risk/high-cholesterol

Mayo Foundation for Medical Education and Research. (2024, June 8). *Mayo Clinic Minute: Poor sleep can be linked to stroke - mayo clinic news network*. Mayo Clinic. https://newsnetwork.mayoclinic.org/discussion/mayo-clinic-minute-poor-sleep-can-be-linked-to-stroke/

Obesity and stroke fact sheet. Obesity Action Coalition. (2023, September 28). https://www.obesityaction.org/resources/obesity-and-stroke-fact-sheet/

Understanding Mental Health and stroke. World Stroke Organization. (n.d.-a). https://www.world-stroke.org/assets/downloads/STROKE_RISK_AND_PREVENTION_LEAFLET_-_MENTAL_HEALTH-EN.pdf

Understanding smoking and stroke. World Stroke Organization. (n.d.). https://www.world-stroke.org/assets/downloads/STROKE_RISK_AND_PREVENTION_LEAFLET_SMOKING-EN.pdf